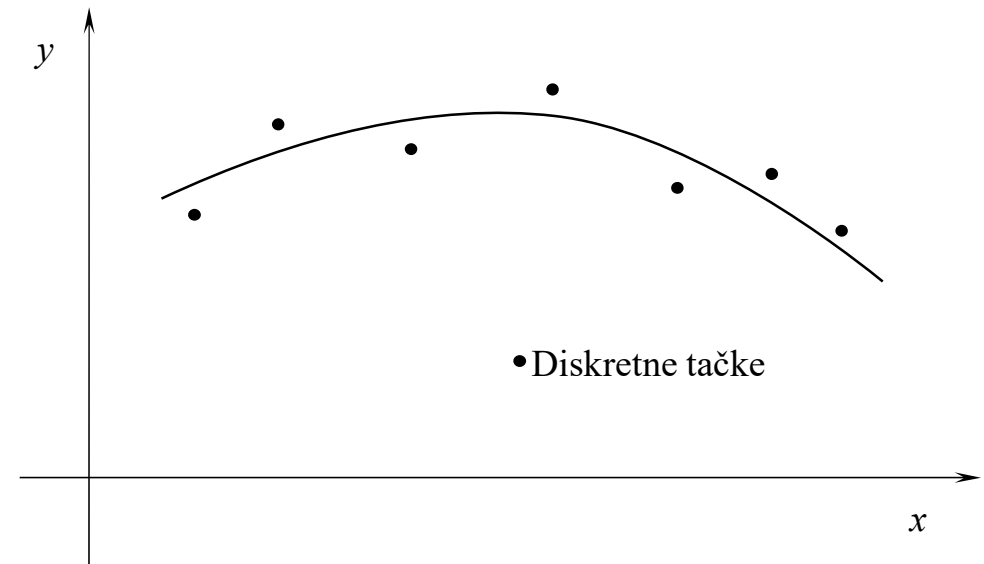
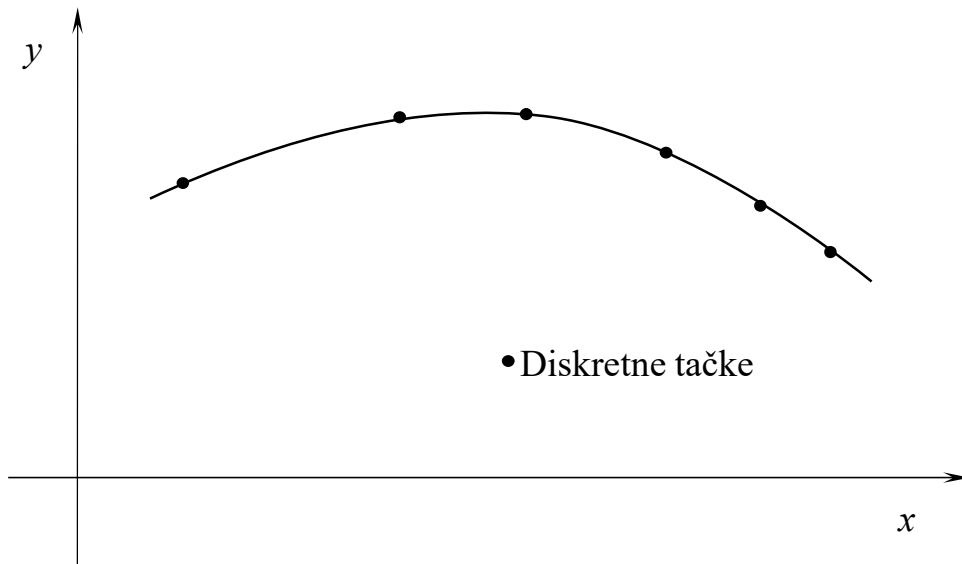


Interpolacija i aproksimacija funkcija

Pristupi u određivanju približnih funkcija

- Interpolacija – tačno poklapanje
- Aproksimacija – približno poklapanje



Interpolacija polinomima

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

- **Weierstrassov aproksimacioni polinom:** Ako je funkcija $f(x)$ neprekidna na intervalu $[a, b]$, tada za svako proizvoljno malo $\varepsilon > 0$ postoji polinom $P_n(x)$, kod kojeg vrijednost n zavisi od vrijednosti ε , tako da za svako x u intervalu $[a, b]$ vrijedi

$$|P_n(x) - f(x)| < \varepsilon$$

- **Teorem o jedinstvenosti rješenja:** Polinom n -tog stepena koji prolazi kroz tačno $n + 1$ diskretnih tačaka je jedinstven.

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Direktna metoda

Problem: Naći interpolacioni polinom n -tog reda koji prolazi kroz $n + 1$ tačaka $[x_i, y_i]$ ($i = 0, 1, \dots, n$)

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$y_0 = a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n$$

$$y_1 = a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n$$

$$y_2 = a_0 + a_1x_2 + a_2x_2^2 + \dots + a_nx_2^n$$

.....

$$y_n = a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n$$

$$D = \begin{vmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{vmatrix} \neq 0$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Primjer

Interpolirati vrijednost za $x = 3.44$, koristeći linearnu, kvadratnu i kubnu interpolaciju, za podatke

x	$f(x)$
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

Napomena: Podaci predstavljaju vrijednosti funkcije $f(x)=1/x$

$$f(3.44)=1/3.44=0.290698$$

• *Linearni polinom (dva podatka) – $y_i = a_0 + a_1 x_i$*

$$\left. \begin{array}{l} 0.294118 = a_0 + a_1 \cdot 3.40 \\ 0.285714 = a_0 + a_1 \cdot 3.50 \end{array} \right\} \Rightarrow \boxed{P_1(x) = 0.579854 - 0.08404x}$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$$f(3.44)=1/3.44=0.290698$$

$$P_1(3.44) = 0.579854 - 0.08404 \cdot 3.44 = 0.290756$$

$$\text{Greška: } f(3.44) - P_1(3.44) = 0.000058$$

x	f(x)
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

•Kvadratni polinom (tri podatka) – $y_i = a_0 + a_1x_i + a_2x_i^2$

$$\left. \begin{array}{l} 0.298507 = a_0 + a_1 \cdot 3.35 + a_2 \cdot 3.35^2 \\ 0.294118 = a_0 + a_1 \cdot 3.40 + a_2 \cdot 3.40^2 \\ 0.285714 = a_0 + a_1 \cdot 3.50 + a_2 \cdot 3.50^2 \end{array} \right\} \Rightarrow P_2(x) = 0.876561 - 0.256080x + 0.0249333x^2$$

$$P_2(3.44) = 0.876561 - 0.256080 \cdot 3.44 + 0.0249333 \cdot 3.44^2 = 0.290697$$

$$\text{Greška: } f(3.44) - P_2(3.44) = -0.000001$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Primjer

• *Kubni polinom (četiri podatka) – $y_i = a_0 + a_1x_i + a_2x_i^2 + a_3x_i^3$*

$$0.298507 = a_0 + a_1 \cdot 3.35 + a_2 \cdot 3.35^2 + a_3 \cdot 3.35^3$$

$$0.294118 = a_0 + a_1 \cdot 3.40 + a_2 \cdot 3.40^2 + a_3 \cdot 3.40^3$$

$$0.285714 = a_0 + a_1 \cdot 3.50 + a_2 \cdot 3.50^2 + a_3 \cdot 3.50^3$$

$$0.277778 = a_0 + a_1 \cdot 3.60 + a_2 \cdot 3.60^2 + a_3 \cdot 3.60^3$$

$$P_3(x) = 1.121066 - 0.470839x + 0.0878x^2 - 0.00613333x^3$$

$$P_3(3.44) = 1.121066 - 0.470839 \cdot 3.44 + 0.0878 \cdot 3.44^2 - 0.00613333 \cdot 3.44^3 = 0.290698$$

$$\text{Greška: } f(3.44) - P_3(3.44) = 0$$

$$f(3.44) = 1/3.44 = 0.290698$$

x	f(x)
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$$f(3.44)=1/3.44=0.290698$$

Lagrangeov interpolacioni polinom

$$P_1(x) = \frac{x-b}{a-b}f(a) + \frac{x-a}{b-a}f(b)$$

$$P_2(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)}f(a) + \frac{(x-a)(x-c)}{(b-a)(b-c)}f(b) + \frac{(x-a)(x-b)}{(c-a)(c-b)}f(c)$$

x	f(x)
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

$$P_n(x) = \frac{(x-b)(x-c)\dots(x-k)}{(a-b)(a-c)\dots(a-k)}f(a) + \frac{(x-a)(x-c)\dots(x-k)}{(b-a)(b-c)\dots(b-k)}f(b) \\ + \dots + \frac{(x-a)(x-b)\dots(x-j)}{(k-a)(k-b)\dots(k-j)}f(k)$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$$P_n(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + \dots + L_n(x)f(x_n) = \sum_{k=0}^n L_k(x)f(x_k)$$

$$L_k(x) = \frac{(x - x_0) \dots (x - x_{k-1})(x - x_{k+1}) \dots (x - x_n)}{(x_k - x_0) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n)} = \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}$$

Primjer

Interpolirati vrijednost za $x = 3.44$, koristeći linearnu, kvadratnu i kubnu interpolaciju, za podatke

x	$f(x)$
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

Napomena: Podaci predstavljaju vrijednosti funkcije $f(x)=1/x$

$$f(3.44)=1/3.44=0.290698$$

$$f(3.44)=1/3.44=0.290698$$

• *Linearni polinom (dva podatka) – $y_i = a_0 + a_1 x_i$*

$$P_1(x) = \frac{x - 3.50}{3.40 - 3.50} 0.294118 + \frac{x - 3.40}{3.50 - 3.40} 0.285714$$

$$P_1(x) = 0.579854 - 0.0840x$$

x	f(x)
3.35	0.298507
3.40	0.294118
3.50	0.285714
3.60	0.277778

• *Kvadratni polinom (tri podatka) – $y_i = a_0 + a_1 x_i + a_2 x_i^2$*

$$P_2(x) = \frac{(x - 3.40)(x - 3.50)}{(3.35 - 3.40)(3.35 - 3.50)} 0.298507 + \frac{(x - 3.35)(x - 3.50)}{(3.40 - 3.35)(3.40 - 3.50)} 0.294118 \\ + \frac{(x - 3.35)(x - 3.40)}{(3.50 - 3.35)(3.50 - 3.40)} 0.285714$$

$$P_2(x) = 0.876561 - 0.256080x + 0.0249333x^2$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Newtonovi interpolacioni polinomi

Prvi Newtonov interpolacioni polinom (za diferenciranje unaprijed)

$$P_n(x) = f_0 + \binom{s}{1} \Delta f_0 + \binom{s}{2} \Delta^2 f_0 + \dots + \binom{s}{n} \Delta^n f_0 = \sum_{k=0}^n \binom{s}{k} \Delta^k f_0$$

$$s = \frac{x - x_0}{h} \quad x = x_0 + sh$$

$$\Delta f_0 = f_1 - f_0$$

$$\Delta^2 f_0 = \Delta f_1 - \Delta f_0 = f_2 - 2f_1 + f_0$$

.....

$$\Delta^n f_0 = \Delta^{n-1} f_1 - \Delta^{n-1} f_0 = \sum_{i=0}^n (-1)^i \binom{n}{i} f_i$$

Tabela podijeljenih razlika

x_i	$f_i^{(0)}$	$f_i^{(1)}$	$f_i^{(2)}$	$f_i^{(3)}$
x_1	$f_1^{(0)}$			
		$f_1^{(1)}$		
x_2	$f_2^{(0)}$		$f_1^{(2)}$	
		$f_2^{(1)}$		$f_1^{(3)}$
x_3	$f_3^{(0)}$		$f_2^{(2)}$	
		$f_3^{(1)}$		
x_4	$f_4^{(0)}$			

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Primjer (prvi *Newtonov* interpolacioni polinom)

Interpolirati vrijednost za $x = 3.44$, koristeći linearnu, kvadratnu

i kubnu interpolaciju, za podatke

x_i	$f_i^{(0)}$	$f_i^{(1)}$	$f_i^{(2)}$	$f_i^{(3)}$	$f_i^{(4)}$
3.1	0.322581				
		-0.010081			
3.2	0.312500		0.000611		
		-0.009470		-0.000053	
3.3	0.303030		0.000558		0.000003
		-0.008912		-0.000050	
3.4	0.294118		0.000508		0.000010
		-0.008404		-0.000040	
3.5	0.285714		0.000468		0.000000
		-0.007936		-0.000040	
3.6	0.277778		0.000428		0.000008
		-0.007508		-0.000032	
3.7	0.270270		0.000396		0.000000
		-0.007112		-0.000032	
3.8	0.263158		0.000364		
		-0.006748			
3.9	0.256410				

$$f(3.44)=1/3.44=0.290698$$

$$P_n(x) = f_0 + \binom{s}{1} \Delta f_0 + \binom{s}{2} \Delta^2 f_0 + \dots + \binom{s}{n} \Delta^n f_0 = \sum_{k=0}^n \binom{s}{k} \Delta^k f_0$$

$$s = \frac{x - x_0}{h} \quad x = x_0 + sh$$

$$s = \frac{x - x_0}{h} = \frac{3.44 - 3.40}{0.1} = 0.4$$

$$P(3.44) = f(3.4) + s \Delta f(3.4) + \frac{s(s-1)}{2} \Delta^2 f(3.4) + \frac{s(s-1)(s-2)}{3!} \Delta^3 f(3.4) + \dots$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$$f(3.44)=1/3.44=0.290698$$

$$P(3.44) = f(3.4) + s \Delta f(3.4) + \frac{s(s-1)}{2} \Delta^2 f(3.4) + \frac{s(s-1)(s-2)}{3!} \Delta^3 f(3.4) + \dots$$

$$P(3.44) = 0.294118 + 0.4 \cdot (-0.008404) + \frac{0.4 \cdot (0.4 - 1)}{2} (0.000468)$$

$$+ \frac{0.4(0.4-1)(0.4-2)}{3!} (-0.0004) + \dots$$

x_i	$f_i^{(0)}$	$f_i^{(1)}$	$f_i^{(2)}$	$f_i^{(3)}$	$f_i^{(4)}$
3.1	0.322581				
		-0.010081			
3.2	0.312500		0.000611		
		-0.009470		-0.000053	
3.3	0.303030		0.000558		0.000003
		-0.008912		-0.000050	
3.4	0.294118		0.000508		0.000010
		-0.008404		-0.000040	
3.5	0.285714		0.000468		0.000000
		-0.007936		-0.000040	
3.6	0.277778		0.000428		0.000008
		-0.007508		-0.000032	
3.7	0.270270		0.000396		0.000000
		-0.007112		-0.000032	
3.8	0.263158		0.000364		
		-0.006748			
3.9	0.256410				

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$$f(3.44)=1/3.44=0.290698$$

$P(3.44) = 0.290756$	linearni	Greška = 0.000058
$= 0.290700$	kvadratni	$= 0.000002$
$= 0.290698$	kubni	$= 0.000000$

Drugi Newtonov interpolacioni polinom (za diferenciranje unazad)

$$P_n(x) = f_0 + \binom{s^+}{1} \Delta f_0 + \binom{s^+}{2} \Delta^2 f_0 + \dots + \binom{s^+}{n} \Delta^n f_0 = \sum_{k=0}^n \binom{s^+}{k} \Delta^k f_0$$

$$\binom{s^+}{i} = \frac{s(s+1)(s+2)\dots[s+(i-1)]}{i!}$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

Primjer (drugi *Newtonov* interpolacioni polinom)

Interpolirati vrijednost za $x = 3.44$, koristeći linearnu, kvadratnu

i kubnu interpolaciju, za podatke

i $x_0=3.5$

Tabela 4.3: Tabela podijeljenih razlika

x_i	$f_i^{(0)}$	$f_i^{(1)}$	$f_i^{(2)}$	$f_i^{(3)}$	$f_i^{(4)}$
3.1	0.322581				
		-0.010081			
3.2	0.312500		0.000611		
		-0.009470		-0.000053	
3.3	0.303030		0.000558		0.000003
		-0.008912		-0.000050	
3.4	0.294118		0.000508		0.000010
		-0.008404		-0.000040	
3.5	0.285714		0.000468		0.000000
		-0.007936		-0.000040	
3.6	0.277778		0.000428		0.000008
		-0.007508		-0.000032	
3.7	0.270270		0.000396		0.000000
		-0.007112		-0.000032	
3.8	0.263158		0.000364		
		-0.006748			
3.9	0.256410				

$$s = \frac{x - x_0}{h}$$

$$s = \frac{x - x_0}{h} = \frac{3.44 - 3.50}{0.1} = -0.6$$

$$\begin{aligned} P(3.44) &= f(3.5) + s\Delta f(3.5) + \frac{s(s+1)}{2!}\Delta^2 f(3.5) \\ &\quad + \frac{s(s+1)(s+2)}{3!}\Delta^3 f(3.5) + \dots \end{aligned}$$

$$\begin{aligned} P(3.44) &= 0.285714 + (-0.6)(-0.008404) \\ &\quad + \frac{(-0.6)(-0.6+1)}{2}(0.000508) \\ &\quad + \frac{(-0.6)(-0.6+1)(-0.4+2)}{3!}(-0.00050) + \dots \end{aligned}$$

Interpolacija i aproksimacija funkcija

Interpolacija polinomima

$P(3.44) = 0.290756$	linearni	Greška = 0.000058
$= 0.290695$	kvadratni	$= -0.000003$
$= 0.290698$	kubni	$= 0.000000$

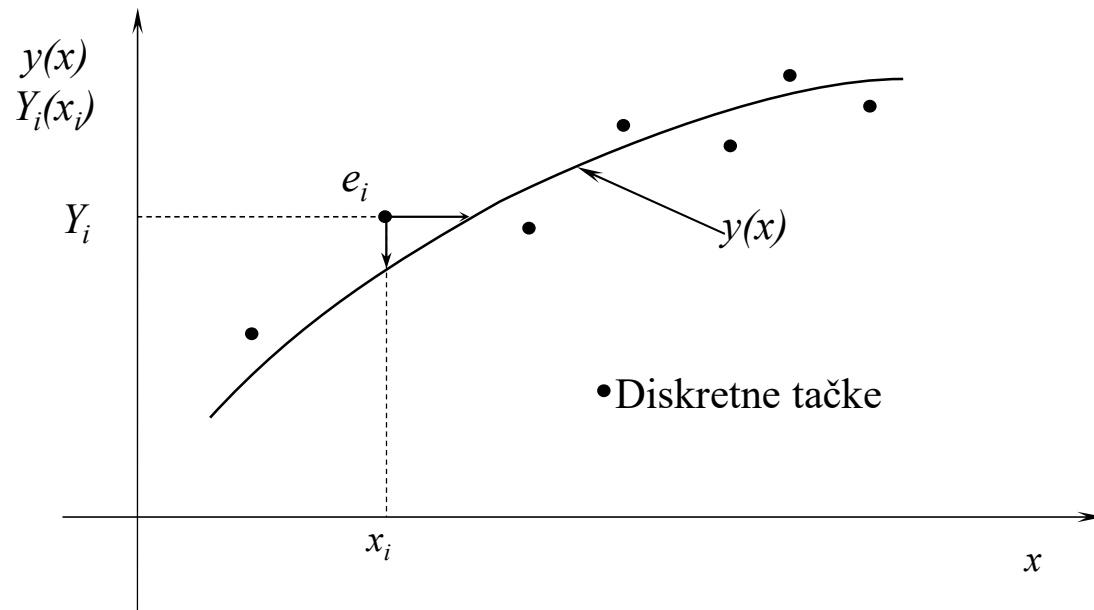
Greška interpolacije

$$R_n(x) = f(x) - P_n(x) = \psi(x) \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

$$\psi(x) = \prod_{i=0}^n (x - x_i) \quad \text{i} \quad \xi \in (a, b)$$

Interpolacija i aproksimacija funkcija

Aproksimacija funkcija



$$e_i = Y_i - y_i$$

Metoda najmanjih kvadrata

$$\min(S)$$

$$S = \sum_{i=1}^N (e_i)^2 = \sum_{i=0}^N (Y_i - y_i)^2$$

Aproksimacija funkcija

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$e_i = Y_i - y_i \qquad S = \sum_{i=1}^N (e_i)^2 = \sum_{i=0}^N (Y_i - y_i)^2$$

$$S(a_0, a_1, a_2, \dots, a_n) = \sum_{i=1}^N (e_i)^2 = \sum_{i=1}^N (Y_i - a_0 - a_1x_i - a_2x_i^2 - \dots a_nx_i^n)^2$$

$$\boxed{\min(S)}$$

$$\frac{\partial S}{\partial a_k} = S(a_0, a_1, a_2, \dots, a_n) = \sum_{i=1}^N 2(Y_i - a_0 - a_1x_i - a_2x_i^2 - \dots a_nx_i^n)(-x_i^k) = 0$$

$$\frac{\partial S}{\partial a_k} = S(a_0, a_1, a_2, \dots, a_n) = \sum_{i=1}^N 2(Y_i - a_0 - a_1 x_i - a_2 x_i^2 - \dots - a_n x_i^n)(-x_i^k) = 0$$

$$a_0 N + a_1 \sum_{i=1}^N x_i + \dots + a_n \sum_{i=1}^N x_i^n = \sum_{i=1}^N Y_i$$

$$a_0 \sum_{i=1}^N x_i + a_1 \sum_{i=1}^N x_i^2 + \dots + a_n \sum_{i=1}^N x_i^{n+1} = \sum_{i=1}^N x_i Y_i$$

.....

$$a_0 \sum_{i=1}^N x_i^n + a_1 \sum_{i=1}^N x_i^{n+1} + \dots + a_n \sum_{i=1}^N x_i^{2n} = \sum_{i=1}^N x_i^n Y_i$$

$$\frac{\partial S}{\partial a_k} = S(a_0, a_1, a_2, \dots, a_n) = \sum_{i=1}^N 2(Y_i - a_0 - a_1 x_i - a_2 x_i^2 - \dots - a_n x_i^n)(-x_i^k) = 0$$

$$aN + b \sum_{i=1}^N x_i = \sum_{i=1}^N Y_i$$

$$a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i Y_i$$

Aproksimacija funkcija

Primjer 1.

Za podatke u tabeli potrebno je odrediti aproksimacioni polinom prvog reda koristeći metodu najmanjih kvadrata u obliku:

$$C_p = a + bT$$

Tabela 4.4: Specifična toplota zraka za niske temperature

$T, \text{ K}$	$C_p, \text{ J/mK} \cdot 10^3$	$\overline{C_p}, \text{ J/mK} \cdot 10^3$	Greška, %
300	1.0045	0.9948	-0.97
400	1.0134	1.0153	0.19
500	1.0296	1.0358	0.61
600	1.0507	1.0564	0.54
700	1.0743	1.0769	0.24
800	1.0984	1.0974	-0.09
900	1.1212	1.1180	-0.29
1000	1.1410	1.1385	-0.22

Interpolacija i aproksimacija funkcija

Aproksimacija funkcija

$$aN + b \sum_{i=1}^N x_i = \sum_{i=1}^N Y_i$$

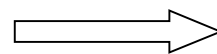
$$a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i Y_i$$

$$8a + b \sum_{i=1}^8 T_i = \sum_{i=1}^8 C_{p,i}$$

$$a \sum_{i=1}^8 T_i + b \sum_{i=1}^8 T_i^2 = \sum_{i=1}^8 T_i C_{p,i}$$

$$8a + 5200b = 8.5331$$

$$5200a + 38000000b = 5632.74$$



$$C_p = 0.933194 + 0.205298 \cdot 10^{-3}T$$

Tabela 4.4: Specifična toplota zraka za niske temperature

$T, \text{ K}$	$C_p, \text{ J/mK} \cdot 10^3$	$\bar{C}_p, \text{ J/mK} \cdot 10^3$	Greška, %
300	1.0045	0.9948	-0.97
400	1.0134	1.0153	0.19
500	1.0296	1.0358	0.61
600	1.0507	1.0564	0.54
700	1.0743	1.0769	0.24
800	1.0984	1.0974	-0.09
900	1.1212	1.1180	-0.29
1000	1.1410	1.1385	-0.22

Primjer 2.

Za podatke u tabeli potrebno je odrediti aproksimacioni polinom drugog reda koristeći metodu najmanjih kvadrata u obliku:

$$C_p = a + bT + cT^2$$

Tabela 4.5: Specifična toplota zraka za visoke temperature

$T, \text{ K}$	$C_p, \text{ J/mK} \cdot 10^3$	$\overline{C_p}, \text{ J/mK} \cdot 10^3$	Greška, %
1000	1.1410	1.1427	0.15
1500	1.2095	1.2059	-0.29
2000	1.2520	1.2522	0.02
2500	1.2782	1.2815	0.26
3000	1.2955	1.2938	-0.13

Interpolacija i aproksimacija funkcija

Aproksimacija funkcija

Tabela 4.5: Specifična toplota zraka za visoke temperature

$T, \text{ K}$	$C_p, \text{ J/mK} \cdot 10^3$	$\overline{C_p}, \text{ J/mK} \cdot 10^3$	Greška, %
1000	1.1410	1.1427	0.15
1500	1.2095	1.2059	-0.29
2000	1.2520	1.2522	0.02
2500	1.2782	1.2815	0.26
3000	1.2955	1.2938	-0.13

$$5a + b \sum_{i=1}^5 T_i + c \sum_{i=1}^5 T_i^2 = \sum_{i=1}^5 C_{p,i}$$

$$a \sum_{i=1}^5 T_i + b \sum_{i=1}^5 T_i^2 + c \sum_{i=1}^5 T_i^3 = \sum_{i=1}^5 T_i C_{p,i}$$

$$a \sum_{i=1}^5 T_i^2 + b \sum_{i=1}^5 T_i^3 + c \sum_{i=1}^5 T_i^4 = \sum_{i=1}^5 T_i^2 C_{p,i}$$

$$5a + 10 \cdot 10^3 b + 22.5 \cdot 10^6 c = 6.1762$$

$$10 \cdot 10^3 a + 22.5 \cdot 10^6 b + 55 \cdot 10^9 c = 12.5413 \cdot 10^3$$

$$22.5 \cdot 10^6 a + 55 \cdot 10^9 b + 142.125 \cdot 10^{12} c = 288.5186 \cdot 10^6$$

$$C_p = 0.965460 + 0.211197 \cdot 10^{-3} T - 0.0339143 \cdot 10^{-6} T^2$$

Nelinearna aproksimacija

Stepena funkcija

$$y = ax^b$$

$$y = ax^b \quad / \ln$$

$$\ln(y) = \ln(a) + b \cdot \ln(x)$$

$$Y = \ln(y), \quad A = \ln(a), \quad X = \ln(x) \quad ; \quad B = b$$

$$Y = A + BX$$

Nelinearna aproksimacija

Eksponencijalna funkcija

$$y = a \cdot e^{bx}$$

$$y = a \cdot e^{bx} \quad / \ln$$

$$\ln(y) = \ln(a) + bx$$

$$Y = \ln(y), \quad A = \ln(a), \quad X = x \quad \text{ i } \quad B = b$$

$$Y = A + BX$$

Nelinearna aproksimacija

Primjer

Za podatke u tabeli potrebno je odrediti nelinearni aproksimacioni polinom oblika

$$y = ae^{bx}$$

x	y	$Y = \ln(y)$
0	3	1.098612
1	6	1.791759
2	12	2.484907
3	24	3.178053
4	48	3.871201

$$y = a \cdot e^{bx}$$

$$y = a \cdot e^{bx} \quad / \ln$$

Interpolacija i aproksimacija funkcija

Nelinearna aproksimacija

x	y	$Y = \ln(y)$
0	3	1.098612
1	6	1.791759
2	12	2.484907
3	24	3.178053
4	48	3.871201

$$y = a \cdot e^{bx}$$

$$y = a \cdot e^{bx} \quad / \ln$$

$$\ln(y) = \ln(a) + bx$$

$$Y = \ln(y), \quad A = \ln(a), \quad X = x \quad \text{ i } \quad B = b$$

$$Y = A + BX$$

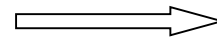
Interpolacija i aproksimacija funkcija

Nelinearna aproksimacija

x	y	$Y = \ln(y)$
0	3	1.098612
1	6	1.791759
2	12	2.484907
3	24	3.178053
4	48	3.871201

$$5A + b \sum_{i=1}^5 x_i = \sum_{i=1}^5 Y_i$$

$$A \sum_{i=1}^5 x_i + b \sum_{i=1}^5 x_i^2 = \sum_{i=1}^5 x_i Y_i$$



$$8A + 10b = 12.424533$$

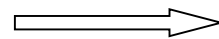
$$10A + 30b = 31.780538$$

$$Y = A + bx = 1.09861 + 0.69315x$$

$$Y = \ln(y), A = \ln(a), X = x \text{ i } B = b$$

$$a = e^A = e^{1.09861} = 3$$

$$b = B = 0.69315$$



$$y = ae^{bx} = 3 \cdot e^{0.69315x} = 3 \cdot (e^{0.69315})^x = 3 \cdot 2^x$$

$$y = 3 \cdot 2^x$$

Numeričko diferenciranje

- Diferenciranje približnim funkcijama
- Formule za diferenciranje

Diferenciranje približnim funkcijama

$$\frac{d}{dx}(f(x)) \cong \frac{d}{dx}(P_n(x))$$

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$f'(x) \cong P'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1}$$

$$f''(x) \cong P''(x) = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2}$$